LOGAN SCHNEIDER

MACHINE LEARNING | DATA SCIENCE | STATISTICS

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PROFESSIONAL SUMMARY

Statistician with an M.S. in Statistics and over three years of model development in cancer research. Co-authored a manuscript on genetic testing in an OB/GYN setting. Currently building a multi-model computer vision system for pickleball coaching that utilizes statistics, analytics, and data visualizations.

TECHNICAL SKILLS

Programming & Data	Python, R/RStudio, SAS, SQL, Git/GitHub
Machine Learning	Predictive modeling, classification, random forests, neural networks, KNN, SVM, NLP
Computer Vision	YOLO, object tracking, Kalman filtering, RTS smoothing, homography, projective geometry, Hough transforms
ML Workflow	Data validation, training/evaluation workflows, error analysis, version control, modular pipeline development, human-in-the-loop validation
Languages	English (native), Russian (fluent)

PROFESSIONAL EXPERIENCE

Biostatistician I Jul 2022 – Oct 2025

Myriad Genetics, Salt Lake City, UT

- Developed statistical models for cancer and genetic-testing research: survival analysis, logistic regression, and mixed-effects models.
- Supported studies from power and sample-size planning through data preparation, modeling, validation, interpretation, and final reporting.
- Developed reproducible R workflows and data-quality checks to identify discrepancies, duplicate records, inconsistent values, and unreliable model inputs before downstream analysis.
- Validated fellow statisticians' workflows, outputs, and interpretations to improve reproducibility, accuracy, and analytical reliability.
- Collaborated with statisticians, researchers, clinicians, and other stakeholders; communicated methods, limitations, and results through reports, visualizations, and presentations.
- Co-authored peer-reviewed research on hereditary-cancer risk assessment and genetic testing in OB/GYN clinics (PubMed).

Statistics Advisor & Graduate Student Instructor Aug 2020 – Aug 2022

Utah State University, Logan, UT

- Advised doctoral research on statistical methods, data visualizations, model interpretation, and reporting findings.
- Investigated systematic bias in national gridded environmental datasets and evaluated approaches for improving predictions in Utah.
- Taught Statistics and Calculus I to classes of 30–50 students, explaining technical concepts to students with varying backgrounds.

MACHINE LEARNING ENGINEERING PROJECT

Pickleball Video Intelligence & Computer Vision Platform | *Independent Project* Jan 2026 – Present

- Designing an end-to-end Python computer-vision system that converts pickleball video into match-analysis data to help players understand their performance.
- Developing a process to refine and improve model detection of the ball, court, and players.
- Building a temporal ball-tracking pipeline combining YOLO detections and Kalman filtering to map ball and player locations from a video onto a pickleball court.
- Refining methods to identify ball contacts, bounces, and trajectories.
- Designed physics-informed trajectory constraints for single-camera footage where image geometry alone cannot uniquely recover real-world ball height.
- Building toward coaching-focused analytics and visualizations that will help players identify and diagnose patterns in shot selection, court and paddle positioning, and strengths and weaknesses of their game. Case study

ADDITIONAL PROJECTS

SNODAS Snow-Water-Equivalent Modeling Jan 2021 – Aug 2022

- Built and published an R package to download and process national gridded environmental data, allow users to create their own spatial-model predictions, and investigate prediction bias across Utah at approximately 1-km resolution. GitHub · Thesis
- Crash severity prediction (random forest, neural network, KNN, SVM) and Python NLP sentiment classification.

Public Health Dashboard 2026

- Client-side dashboard (JavaScript, Plotly.js) fetching live CDC respiratory-illness and openFDA adverse-event data with caching and error handling, no backend. In progress. Preview

Customer Churn Prediction 2026

- End-to-end churn analysis (Python, XGBoost, SHAP): 0.85 ROC-AUC, VIF-diagnosed multicollinearity in the baseline, and a cost-based retention framework estimating $1.63M in revenue at risk under stated assumptions.

EDUCATION

M.S. in Statistics, Utah State University 2022

Thesis: An Ensemble Approach for Mapping Snow Water Equivalent in Utah (published 2023)

B.S. in Mathematics/Statistics, Minor in Russian, Utah State University 2019